Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 18.3

$$\int x^3 \sin x dx$$

$$\left\{\begin{array}{l} u = x^3 \\ dv = \sin x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = 3x^2 dx \\ v = -\cos x \end{array}\right.$$

$$\int x^3 \sin x dx = -x^3 \cos x + \int 3x^2 \cos x dx$$

$$= -x^3 \cos x + 3 \int x^2 \cos x dx$$

$$\left\{\begin{array}{l} u = x^2 \\ dv = \cos x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = 2x dx \\ v = \sin x \end{array}\right.$$

$$3 \int x^2 \cos x dx = 3 \left(x^2 \sin x - \int 2x \sin x dx \right)$$

$$= 3x^2 \sin x - 6 \int x \sin x dx$$

$$\left\{\begin{array}{l} u = x \\ dv = \sin x dx \end{array}\right. \Rightarrow \left\{\begin{array}{l} du = dx \\ v = -\cos x \end{array}\right.$$

$$\int x \sin x dx = -x \cos x + \int \cos x dx$$

$$= -x \cos x + \sin x$$

$$\int x^3 \sin x dx = -x^3 \cos x + 3x^2 \sin x - 6 \left(-x \cos x + \sin x \right)$$

$$= -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$$

References